LAB WORKBOOK

22SDCS03A/22SDCS03R
JAVA FULL STACK DEVELOPMENT + MICROSERVICES

TEAM JFSD
K L UNIVERSITY | VADDESWARAM

LAB WORKBOOK

22SDCS03A/22SDCS03R – JFSD + MICROSERVICES

STUDENT NAME	
STUDENT ID	
YEAR	
SEMESTER	
SECTION	
FACULTY NAME	

DEPARTMENT VISION AND MISSION

Vision

To be a Department of International Repute through Continuous Research, Innovation and Industry Led Curriculum.

Mission

To Impart Quality Education with Social Consciousness and make them globally competent.

Program Educational Objectives

1. Practice engineering in a broad range of industrial, societal and real-world applications.
2. Pursue advanced education, research and development, by adapting creative and innovative practices in their professional careers.
3. Conduct themselves in a responsible, professional, and ethical manner.
4. Participate as leaders in their fields of expertise and in activities that support service and economic development throughout the world.

PROGRAM OUTCOMES		
PO	**Graduate Attributes**	**Program Outcome Description**
1	**Engineering Knowledge**	To impart mathematics, science, & engineering knowledgeto develop skills to solve complex engineering problems.
2	**Problem Analysis**	Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
3	**Design/ development of solutions**	Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
4	**Conduct investigations of complexproblems**	An ability to use research-based knowledge and research methods including design of experiments, analysis and interpretation of data and synthesis of the information to provide valid conclusions.
5	**Modern tool usage**	Ability to create, select and apply appropriate techniques, resources and modern engineering activities, while understanding its limitations.
6	**The engineer and society**	Ability to apply reasoning and the contextual knowledge toassess social & health, safety, legal and cultural issues andthe consequent responsibilities relevant to the professionalengineering practices.
7	**Environment and sustainability**	Ability to demonstrate the engineering knowledge to find solutions to contemporary issues by understanding their impact on societal and environmental contexts, towards sustainable development
8	**Ethics**	An ability to apply ethical principles and commit to professional ethics and responsibilities and norms of engineering practice.
9	**Individual and teamwork**	To inculcate abilities to be able to act as a leader as well as team player effectively in multi-disciplinary settings
10	**Communication**	To develop oral and written communication skills to articulate the complex engineering activities with the engineering community and and society effectively through reports and design documentation, make effective presentations, and give and receive clear instructions.
11	**Project management and finance**	To develop working knowledge and understanding of the engineering and management principles to manage projects in multi-disciplinary environments.
12	**Lifelong learning**	To inculcate the habit of constant knowledge upgrading habit to meet the ever-changing technology and industry needs.
PROGRAM SPECIFIC OUTCOMES		
PSO1	An ability to design and develop software projects as well as to analyze and test user requirements.	
PSO2	Working knowledge on emerging technologies as per the industry requirements	

COURSE COORDINATOR

Mr. J. SURYA KIRAN

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2024-25 ODD SEMESTER JFSD LAB CONTINUOUS EVALUATION

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1						
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#LAB – 1 ➔ Hibernate CRUD Operations

Prerequisites:

- **Basic Idea on ORM, JPA and Hibernate API**

Rama, working as Assistant Professor at K L University, needs to keep up the information about his students in his database. He wants to save student details such as Id, name, gender, department, program, date of birth, contact number, graduation status, CGPA, no of backlogs.

Help him to develop a hibernate maven project to perform following operations

1. Insert the details of student graduated. The application should ask the user whether he wants to insert a detail of a student each time until he says 'no'. For every 'yes' it should gather the details of the student i.e., Id, name, gender, department, program, date of birth, contact number, graduation status, CGPA, no of backlogs.
2. Fetch a student record by asking student id as input
3. Update specific student record by providing student id and details to be updated.
4. Delete specific student record based on student id.

VIVA QUESTIONS:

1. What is ORM and give some examples for ORM tools?
2. What are the advantages of hibernates over JDBC?
3. What is lazy loading in hibernate?
4. What are the core components of Hibernate?
5. What are the different methods to be performed using Persistence Object.

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#LAB – 2 ➔ HQL & HCQL

Prerequisites:

- **Basic Idea on Hibernate API**
- **Basic Knowledge on HQL and HCQL to perform 'CRUD' Operation**

Perform the following tasks taking reference from Lab 1 with Hibernate Query Language (HQL) and Hibernate Criteria Query Language (HCQL).

1. HQL query to display all student records with all columns.
2. HQL query to display all student records with specific columns.
3. HQL query to display only names of all students whose CGPA is greater than 7.
4. HQL query to delete a Student whose Student ID is 30101 using input parameter (?) as well as named parameter (:)
5. HQL query to update few details of a student whose Student ID is 30102 using input parameter (?) as well as named parameter (:)
6. HQL query to perform aggregate functions like count (*), sum(), avg(), min(), max() functions on CGPA column.
7. HCQL query to display specific columns from set of student records.
8. HCQL query to get 5th to 10th records.
9. HCQL query to apply greater than, less than, greater than equal to, less than equal to, equal to and not equal to on CGPA column.
10. HCQL query to get the records in ascending as well as descending order based on Student Name.

VIVA QUESTIONS:

1. What is HQL? What is the difference between SQL and HQL?
2. What is criteria API?
3. Difference between named parameter and position parameter
4. Advantages of HQL
5. Does Hibernate support Native SQL Queries?

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#LAB – 3 ➔ Hibernate Inheritance Mapping

Prerequisites:

- **Basic Idea on Hibernate and Java Inheritance**
- **Basic Idea on Generator Classes**

1. A Library System wants to manage its inventory through inheritance using the Table per Class Hierarchy strategy. Create a base class named **Item** with attributes **itemId** and **title**. Derive two classes from it: **Book** containing **author** and **isbn**, and **DVD** containing **director** and **regionCode**. When an object of a derived class is saved, the object of the base class should also get stored in the database in a single table. Use a discriminator column to identify the class type. Apply the concept of Table per Class Hierarchy through the Hibernate Framework.
2. Mr. John Doe is a Fleet Manager at a company. Create a base class **Vehicle** with attributes **vehicleId** and **model**. Derive two classes: **Car** containing **seats** and **fuelType**, and **Bike** containing **type** and **engineCapacity**. Each derived class should be stored in its own table, containing the attributes of both the base and derived classes. Use the concept of Table per Concrete Class Hierarchy through the Hibernate Framework and observe the structure of data stored in the database.
3. An Online Store wants to manage its product catalog using inheritance through the Table per Subclass Hierarchy strategy. Create a base class named **Product** with attributes **productId** and **name**. Derive two classes from it: **Electronics** containing **warranty** and **brand**, and **Clothing** containing **size** and **material**. Each class should be stored in its own table, with a primary key that is also a foreign key to the base class table. Apply the concept of Table per Subclass Hierarchy using the Hibernate Framework and observe how data is stored in the database.

VIVA QUESTIONS:

1. What is Inheritance Mapping and List out Inheritance Mapping Hierarchy?
2. Explain each Type of Hierarchy.

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#LAB – 4 ➔ Spring Framework with DI & IoC

Prerequisites:

- **General Idea on Spring Framework**
- **Basic Idea on DI & IoC**

1. You are developing a basic student grading system. Write a Java program that demonstrates the use of both primitive and non-primitive data types. Create a student class with fields for Student ID (as String), Name (as String), Gender (as String), Age (as int), and Date of Birth (as Date). Implement a constructor and setter dependency injection to initialize these fields then display the student's details in the main method using beans.
2. Set up a simple Spring application to demonstrate dependency injection and autowiring. Define a Vehicle interface with a move() method, and implement two classes, Bike and Car, each providing its own implementation of the move() method. Create a journey class that has a Vehicle field, which should be autowired by Spring. Configure your Spring application context using either XML or Annotations to support autowiring. Finally, write a main method or test case to start a journey, ensuring that a Vehicle implementation (either Bike or Car) is injected, and call the move() method to observe the behavior of different vehicle types.

VIVA QUESTIONS:

1. Different modules in Spring Architecture
2. Dependency Injection and types
3. Spring IoC Container and types
4. Advantages of Spring Framework

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#LAB – 5 ➔ Spring Web MVC Demo & Annotations

Prerequisites:

- **General Idea on Spring Framework**
- **Modules of Spring Framework**
- **General Idea on Spring Web MVC**

1. Create a Spring MVC controller class named **ClientController** that demonstrates various request mappings and handling methods. The controller should include the following demo operations:
   - Create a method that maps to the root URL ("/") and returns a ModelAndView object with the view name set to "home".
   - Create a method that maps to the URL "/demo1" using @GetMapping and returns a ModelAndView object with the view name set to "admin".
   - Create a method that maps to the URL "/demo2" using @GetMapping, sets the view name to "user", and adds an attribute "name" with the value "klu" to the ModelAndView.
   - Create a method that maps to the URL "/demo3" using @GetMapping, sets the view name to "about", and adds an attribute "msg" with the value "Hello World" to the ModelAndView.
   - Create a method that maps to the URL "/demo4" using @GetMapping and returns a string "JFSD" as the HTTP response body.
   - Create a method that maps to the URL "/demo5" using @GetMapping, accepts two request parameters "a" and "b", and returns their sum as a string in the HTTP response body.
   - Create a method that maps to the URL "/demo6/{fname}/{lname}" using @GetMapping, accepts path variables "fname" and "lname", sets the view name to "demo", and adds the concatenated full name as an attribute "name" to the ModelAndView.

VIVA QUESTIONS:

1. What is the role of the DispatcherServlet in Spring MVC?
2. How does Spring MVC handle form submission?
3. Explain the concept of view resolvers in Spring MVC.
4. What is the difference between @Controller and @RestController in Spring MVC?
5. How can you handle exceptions in a Spring MVC application?

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#LAB – 6 ➔ Spring Web MVC CRUD Operations with Database Integration

Prerequisites:

- **General Idea on Spring Web MVC and Form Handling**
- **Hibernate Template**

1. Almost all the students from various colleges/Universities are aware of our National Level Techno- Management Fest "Samyak". This Year Samyak organizers are planning to conduct an event so that other college students can also participate in it. For that purpose, the Samyak Chief needs a spring MVC application to maintain the details of the participants. He/she who is registering for the event must be currently pursuing a bachelor's degree. A list of events should be provided so that the user can opt based on his/her choice.
    A. User must register first to get credentials for login.
    B. Admin must view all registered users along with the count.
    C. Admin can add/update/delete/view events.
    D. After User login, he/she can register into any number of events after payment.
    E. Admin must view list of members registered for a particular event.

VIVA QUESTIONS:

1. What are the advantages of using Spring MVC?
2. What are the modules in Spring Framework?
3. How can beans be made singleton or prototype?
4. What are the advantages of DAO module?
5. Mention the types of IOC?

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#LAB – 7 ➔ Spring Boot Web MVC Demo & Annotations

Prerequisites:

- **General Idea on Spring Boot MVC Architecture**

1. Create a Spring Boot Web MVC application that demonstrates various request mappings and handling methods using a **customer** object. Implement the following demo operations
   - Create a method that maps to the URL "/demo1" and returns an integer.
   - Create a method that maps to the URL "/demo2" and returns a double value
   - Create a method that maps to the URL "/demo3" using @GetMapping and returns a formatted HTML string.
   - Create a method that maps to the URL "/demo4" using @GetMapping, initializes a string "KLEF", and returns a concatenated string "I Study at " with the initialized string.
   - Create a method that maps to the URL "/demo5/{id}" using @GetMapping, accepts a path variable "id", and returns its value.
   - Create a method that maps to the URL "/demo6/{a}/{b}" using @GetMapping, accepts two path variables "a" and "b", and returns their sum as a string.
   - Create a method that maps to the URL "/demo7" using @GetMapping, accepts a request parameter "id", and returns it as a string.
   - Create a method that maps to the URL "/demo8/{name}" using @GetMapping, accepts a path variable "name", and returns it as a string.
   - Create a method that maps to the URL "/addcustomer" using @PostMapping, accepts a Customer object in the request body, adds it to a list, and returns a confirmation message "Customer Added Successfully".
   - Create a method that maps to the URL "/viewcustomer" using @GetMapping and returns the list of all Customer objects added.

VIVA QUESTIONS:

1. What are the key differences between Spring MVC and Spring Boot MVC?
2. How does Spring Boot simplify the development of Spring applications?
3. Explain how to use @RequestBody and @ResponseBody annotations in Spring Boot.
4. explain the role of the @Autowired annotation in Spring Boot and how it works in the context of Dependency Injection?
5. What is the purpose of application.properties or application.yml in a Spring Boot application?

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#LAB – 8 ➔ Spring Boot with REST API and Database Integration

Prerequisites:

- **Basic Idea on Spring Boot and Spring Data JPA**
- **Basic Idea on Webservices (Rest API)**

1. We KLUian's are very familiar with various kinds of Hackathons that are being conducted in our university. We also know that organizing such kind of activities is a bit of tough task for a Mundane. Your friend who is part of organizing these Hackathons wants to develop a Spring Boot Application using RESTful API to manage all the things. He wants you to help him to build an application such that he can manage the student details who are participating in Hackathon.

   Requirements for the Hackathon Application:
   a. **GET -** All Student's and Particular Student (By ID)
   b. **POST -** Add a student.
   c. **PUT -** Update a Student by ID
   d. **DELETE -** Delete a Student by ID

VIVA QUESTIONS:

1. What is the role of Actuator in Spring Boot?
2. What are the features of RESTful Webservices?
3. What is meant by statelessness in REST?
4. What is Auto Configuration in Spring Boot?
5. What is Spring Boot Initializer? What are the Advantages of it?

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#LAB - 9 ➔ Spring Boot Web MVC CRUD Operations with Database Integration

Prerequisites:

- **General Idea on Spring Boot MVC and Form Handling**
- **General Idea on Spring Data JPA**

Develop a Spring Boot web application to manage a list of products in a warehouse. The application should handle CRUD operations to manage product details such as Product ID, Name, Description, Price, and Quantity. The application should include features to add new products, display a list of all products, update existing product details, and delete products from the database. Use Spring Web MVC for handling HTTP requests, Spring Data JPA for database interactions. Ensure the application is configured to connect to a MySQL/PostgreSQL database and implement both setter-based or constructor-based dependency injections to manage service and repository layers effectively.

VIVA QUESTIONS:

1. What is the role of Actuator in Spring Boot?
2. What are the advantages of Spring Boot?
3. What are the Spring Boot Features?
4. Differences between Spring and Spring Boot?
5. What are the Basic Annotations in Spring Boot?

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#LAB - 10 ➔ Microservices using Spring Boot - 1

Prerequisites:

- **Basic Idea in Spring Boot**
- **Basic Idea on Microservices**

1. Design a Java-based microservice architecture to help Raj, a B. Tech (CSE) student at KLU to choose his courses.
   - **Student Service**: A Microservice to manage student information.
   - **Course Service**: A Microservice to manage courses.
   - **Course Registration Service**: A Microservice to handle course registration and track the courses selected by students.

   I. **Student Service**:
      a) Add a new student.
      b) View the list of all students.

   II. **Course Service**:
      a) Add a new course.
      b) View the list of all courses.

   III. **Course Registration Service**:
      a) Register a student for a course.
      b) View the list of courses a student is registered for.

VIVA QUESTIONS:

1. What are the Advantages of Microservices?
2. State some technologies commonly used to implement microservices?
3. What are the different types of Microservices?
4. What are the characteristics of Microservices?
5. What are the main components in Microservices?

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#LAB – 11 ➔ Microservices using Spring Boot - 2

Prerequisites:

- **Basic Idea in Spring Boot**
- **Basic Idea on Microservices**

1. In continuation of the above requirements, now implement the update and delete operations for each of the three services.
   i. **Student Service**:
      a) Update student information.
      b) Delete a student
   ii. **Course Service**:
      a) Update Course Information.
      b) Delete a course.
   iii. **Course Registration Service**:
      a) Update a student's course registration.
      b) Delete a student's course registration.

VIVA QUESTIONS:

1. How can we access RESTful in Microservices?
2. What's the difference between a microservices-oriented architecture (MOA) and a service-oriented architecture (SOA)?
3. What does the term "bounded context" mean in relation to microservices?
4. Explain three types of Tests for Microservices?
5. How is distributed tracing used in Microservices?

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#LAB - 12 ➔ Spring Cloud with Eureka Client, Server and API Gateway

Prerequisites:

- **Basic Idea on Spring Cloud**
- **Basic Idea on Netflix Eureka Client & Server**

1. Extend the previously developed microservice architecture for KL University's online registration system to utilize Eureka for service discovery and showcase the services in the Eureka client with specific port numbers.

   a) Student Service:

   i) Continue using the Student Service API to manage student information.

   b) Course Service:

   i) Continue using the Course Service API to manage courses.

   c) Course Registration Service:

   i) Continue using the Course Registration Service API to handle the registration process and track the courses and sections selected by students.

❖ Use Spring Cloud Eureka for service discovery.

❖ Ensure all services are properly registered and visible in the Eureka client, demonstrating the microservice architecture with specific port numbers.

❖ Implement API Gateway to implement Single Port numbers which will be directed individual Eureka Servers based on patterns.

VIVA QUESTIONS:

1. What are the advantages of using Spring Cloud?
2. What is the difference between Spring Boot and Spring Cloud?
3. What does one mean by Load Balancing?
4. What does service registration and discovery mean? How is Spring Cloud implemented?
5. What is API Gateway & Uses ?

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#LAB - 13 ➔ Spring boot unit testing with Junit Framework

Prerequisites:

- **Basic Idea on Spring Boot**
- **Basic Idea on Junit Framework**

Develop a Spring Boot application to manage a list of employees in an organization. The application should support CRUD operations to manage employee details such as Employee ID, Name, Designation, Department, Salary, and Contact Information. Implement comprehensive unit tests for each component of the application using the JUnit framework. Specifically, create tests for the service layer to verify the business logic, the repository layer to ensure correct database interactions, and the controller layer to validate HTTP request handling. Ensure high code coverage and proper validation of each functionality using JUnit assertions and annotations.

VIVA QUESTIONS:

1. What is the purpose of @WebMvcTest annotation in Spring Boot?
2. Explain different annotations in Junit Framework.
3. How can you test RESTful endpoints in a Spring Boot application?
4. What are the advantages of using JUnit for unit testing in Spring Boot?

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#LAB - 14 ➔ Spring Boot Security with JWT Token for RBAC

Prerequisites:

- **Basic Idea on Spring Security**
- **Basic Idea on JWT Tokens & RBAC**

1. An e-commerce platform requires secure access to different functionalities based on user roles. There are two main roles: USER and ADMIN. Users can view products and their details, while admins can add, update, and delete products. Implement JWT-based authentication to secure the application and ensure that only authorized users can access specific endpoints.
2. Develop a Spring Boot application with JWT-based security for role-based authentication and authorization. Create unit tests using JUnit to ensure the security mechanisms are correctly implemented.

- Create a User Model
- Create a Product Model
- Create User and Product Repositories
- Implement JWT Utility Class
- Configure Spring Security
- Create Authentication Controller
- Create Product Controller

VIVA QUESTIONS:

1. Explain the concept of JWT and its role in securing web applications.
2. Describe the process of setting up JWT-based authentication in a Spring Boot application.
3. What is the significance of role-based authorization, and how is it implemented in Spring Security?
4. How do you create and configure security filters for handling JWT tokens in Spring Boot?
5. Discuss the use of @PreAuthorize and @Secured annotations in role-based access control.

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DEPARTMENT OF CSE
COURSE CODE: 22SDCS03A/22SDCS03R
JAVA FULL STACK DEVELOPMENT + MICROSERVICES

Date of the Session: __/__/____ **Time of The Session: _____to_____**

#LAB - 15 ➔ Spring Boot with ThymeLeaf Library

Prerequisites:

- **Basic Idea on Spring Boot**
- **Basic Idea on ThymeLeaf Library**

1. A library wants to create a web application for managing books. The application should allow users to view a list of books, add a new book, update book details, and delete a book. ThymeLeaf will be used for rendering HTML pages dynamically based on the data provided by the Spring Boot backend.
2. Develop a Spring Boot application integrated with ThymeLeaf to manage book information.

VIVA QUESTIONS:

1. Explain the role of ThymeLeaf in a Spring Boot application.
2. Describe the process of setting up ThymeLeaf with Spring Boot.
3. How do you create and configure a ThymeLeaf template in a Spring Boot application ?
4. Discuss the use of ThymeLeaf expressions for dynamic content rendering in HTML templates.
5. How do you handle form submissions in Spring Boot with ThymeLeaf ?

(For Evaluator's use only)

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